

A-5.  $x^2 \cdot y' + y = 1 \Rightarrow y' + \underbrace{\frac{1}{x^2}}_{p(x)} \cdot y = \underbrace{\frac{1}{x^2}}_{q(x)}$  Linear dif. denk.

$$y = e^{-\int p(x) dx} \cdot [c + \int q(x) e^{\int p(x) dx} dx]$$

$$y = e^{-\int \frac{dx}{x^2}} \cdot [c + \int \frac{1}{x^2} \cdot e^{\int \frac{dx}{x^2}} dx]$$

$$y = e^{+\frac{1}{x}} \cdot [c + \underbrace{\int \frac{1}{x^2} \cdot e^{-\frac{1}{x}} dx}_A]$$

$$A = \int e^{-\frac{1}{x}} \cdot \frac{dx}{x^2} \Rightarrow -\frac{1}{x} = u \Rightarrow \frac{1}{x^2} dx = du \text{ olur.}$$

$$A = \int e^u du = e^u \text{ yani } A = e^{-\frac{1}{x}} \text{ olur}$$

$$y = e^{\frac{1}{x}} \cdot [c + e^{-\frac{1}{x}}] = ce^{1/x} + 1 \text{ elde edilir.}$$